

# Farzad Ahmed

Ph.D. Student, Information Technology | Department of Information Sciences and Technology, George Mason University

4400 University Drive, Fairfax, VA 22030, USA | 571-544-2973 | farzadahmed6@gmail.com

farzad-1996.github.io | scholar.google.com/citations?user=tFYmHMsAAAAJ | github.com/Farzad-1996 | orcid.org/0009-0003-2030-6592

## RESEARCH INTERESTS

Agentic AI and multi-agent large language model architectures; retrieval-augmented generation; evaluation and benchmarking of large language models; natural language processing; health informatics and clinical NLP; explainable AI.

## EDUCATION

**Ph.D. in Information Technology**, George Mason University — Fairfax, VA, USA

Fall 2023 – Present

Concentration: Information Sciences and Technology, College of Engineering and Computing. Expected Spring 2028.

Advisor: Özlem Uzuner, Professor and Chair, Department of Information Sciences and Technology.

**M.S. in Computer Science**, George Mason University — Fairfax, VA, USA | CGPA: 3.90 / 4.00

Awarded Aug 2026

**B.Sc. in Computer Science and Engineering**, Ahsanullah University of Science and Technology — Dhaka, Bangladesh

2015 – 2019

CGPA: 3.923 / 4.00, 1st in Merit Position. Thesis: "Early Depression Detection from Social Media Using Deep Learning Techniques," advised by Faisal Muhammad Shah.

## RESEARCH EXPERIENCE

**Graduate Research Assistant**, MasonNLP Research Group, George Mason University — Fairfax, VA

Aug 2024 – Present

- Designed a multi-agent large language model architecture: five specialist reasoning agents across a LangGraph DAG with sequential context dependencies, adjudicated by a coordinator constrained to cite only agent-flagged evidence. Benchmarked across 925 clinical notes at roughly 5,500 model calls per run.
- Extended the system with retrieval-augmented generation over a 2,080-chunk FAISS vector index built from 447 FDA drug labels ingested through the OpenFDA and RxNorm APIs, measuring whether external grounding closes the failure modes of purely parametric reasoning.
- Benchmarked open-weight against proprietary foundation models under zero-shot, few-shot, and context-aware prompting, and against LoRA fine-tuned transformer encoders trained on HPC GPU clusters with distributed multi-node, multi-GPU jobs.
- Co-led an LLM-driven ETL framework converting heterogeneous EHR databases into the OMOP Common Data Model, benchmarking seven models across MIMIC-IV, OpenMRS, and Synthea on executability, content fidelity, and referential integrity.
- Ran controlled ablations with one variable per condition, paired McNemar exact significance testing, and Wilson confidence intervals; calibrated performance against human inter-annotator agreement and reported null results where effects did not clear the noise floor.
- Collaborated with radiologists, clinical informaticians, and public health researchers across two institutions, co-authoring four manuscripts in two years.

## TEACHING EXPERIENCE

**Lecturer**, Department of CSE, Ahsanullah University of Science and Technology — Dhaka, Bangladesh

Dec 2020 – Aug 2023

- Taught undergraduate computer science and programming in Python and C++ — data structures, algorithms, debugging, and software testing practices.
- Supervised student research teams through problem scoping, experiment design, and evaluation, mentoring them to six peer-reviewed publications.

**Graduate Teaching Assistant**, George Mason University — Fairfax, VA, USA

Aug 2023 – Aug 2024

- Supported graduate computer science coursework, leading debugging sessions and evaluating student code across a large enrollment.

**Contractual Lecturer**, Department of CSE, BRAC University — Dhaka, Bangladesh

Feb 2020 – Dec 2020

- Delivered undergraduate computer science instruction while conducting deep learning research on early depression detection from social media text.

## JOURNAL PUBLICATIONS

- Park, N., **Ahmed, F.**, Sun, Z., Lybarger, K., et al. "Automated Identification of Incidentalomas Requiring Follow-Up: A Multi-Anatomy Evaluation of LLM-Based and Supervised Approaches." *Journal of Biomedical Informatics* (Elsevier), 2026.
- Shawon, M. T. R., Shibli, G. M. S., **Ahmed, F.**, Joy, S. K. S. "Explainable Cost-Sensitive Deep Neural Networks for Brain Tumor Detection from Brain MRI Images Considering Data Imbalance." *Multimedia Tools and Applications* (Springer), 84(35): 43615–43642, 2025.
- Shah, F. M., Joy, S. K. S., **Ahmed, F.**, Hossain, T., Humaira, M., Ami, A. S., Paul, S., Jim, M. A. R. K., Ahmed, S. "A Comprehensive Survey of COVID-19 Detection Using Medical Images." *SN Computer Science* (Springer), 2: 434, 2021.

## CONFERENCE PUBLICATIONS

- Afrin, N., Hasan, M. M., Safin, M. F. E., Amin, K. R., Haque, M. Z., **Ahmed, F.**, Shawon, M. T. R. "Bengali License Plate Recognition: Unveiling Clarity with CNN and GFP-GAN." *26th International Conference on Computer and Information Technology (ICCIT)*, 2023.
- Joy, S. K. S., **Ahmed, F.**, Mahamud, A. H., Mandal, N. C. "An Empirical Study on How the Developers Discussed About Pandas Topics." *International Conference on Machine Intelligence and Emerging Technologies (MIET)*, LNICST, Springer, 2022.
- Mouri, A. G., Talukder, P., Anik, T. R., Rahman, I. S. I., Joy, S. K. S., Shawon, M. T. R., **Ahmed, F.**, Mandal, N. C. "An Empirical Study on Bengali News Headline Categorization Leveraging Different Machine Learning Techniques." *25th International Conference on Computer and Information Technology (ICCIT)*, 2022.

- [7] **Ahmed, F.**, Sultana, S., Reza, M. T., Joy, S. K. S., Alam, M. G. R. "Interpretable Movie Review Analysis Using Machine Learning and Transformer Models Leveraging XAI." *IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, 2022.
- [8] Reza, M. T., **Ahmed, F.**, Sharar, S., Rasel, A. A. "Interpretable Retinal Disease Classification from OCT Images Using Deep Neural Network and Explainable AI." *International Conference on Electronics, Communications and Information Technology (ICECIT)*, 2021.
- [9] Shah, F. M., **Ahmed, F.**, Joy, S. K. S., Ahmed, S., Sadek, S., Shil, R., Kabir, M. H. "Early Depression Detection from Social Network Using Deep Learning Techniques." *2020 IEEE Region 10 Symposium (TENSYP)*, 2020.

---

## MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

- [10] **Ahmed, F.**, Jerome, J. A., Yetisgen, M., Uzuner, Ö. "A Systematic Analysis of Large Language Models with RAG-enabled Dynamic Prompting for Medical Error Detection and Correction." Under review, *JMIR Medical Informatics*.
- [11] **Ahmed, F.**, Saelens, B. E., Yetisgen, M., Uzuner, Ö. "Cascade Prompting for Automated Document Classification: A Comparative Evaluation of Open-Weight and Proprietary Large Language Models." Under review, *Journal of Clinical and Translational Science*.
- [12] LLM-driven ETL for converting heterogeneous EHR databases to the OMOP Common Data Model. Manuscript in preparation for the *Journal of Biomedical Informatics*.

---

## TECHNICAL SKILLS

**Agentic AI & LLMs:** Multi-agent orchestration (LangGraph, LangChain), agent coordination and constrained adjudication, LLM APIs (OpenAI, Azure OpenAI Service, AWS Bedrock), prompt engineering, chain-of-thought reasoning, schema-constrained generation (Pydantic), LLM evaluation and benchmarking

**Retrieval & Vector Search:** Retrieval-augmented generation (RAG), FAISS vector indexing, text embeddings, chunking strategy and retrieval optimization

**Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, transformer encoders (BERT-family), GPT, LLaMA, LoRA parameter-efficient fine-tuning, CNNs, Vision Transformers (ViT), OpenCV, distributed multi-node and multi-GPU training

**Explainable & Responsible AI:** SHAP, LIME, Grad-CAM, evidence-constrained model outputs, cost-sensitive learning under severe class imbalance

**Evaluation & Statistics:** Controlled ablation design, paired significance testing (McNemar exact), Wilson confidence intervals, inter-annotator agreement, precision/recall analysis, held-out evaluation

**Health Informatics Data:** OMOP Common Data Model, OHDSI Athena vocabularies, MIMIC-IV, OpenMRS, Synthea, OpenFDA and RxNorm APIs, clinical text and radiology report processing

**Programming & Systems:** Python (primary, 7 years — pandas, NumPy), SQL, Java, C, C++, C#, JavaScript, Unix/Linux, REST APIs, DAG orchestration, thread-pool concurrency, checkpoint/resume durability, Git and GitHub, unit and integration testing, HPC GPU clusters

---

## HONORS AND AWARDS

- George Mason University Graduate Award — merit-based award for excellence in graduate research.
- Third Place, Student Research Presentations — Inaugural Ph.D. Symposium, Department of Information Sciences and Technology, George Mason University, Spring 2025.
- Dean's List of Honors, Ahsanullah University of Science and Technology — sustained academic excellence during undergraduate studies.
- 1st in Merit Position, B.Sc. in Computer Science and Engineering, Ahsanullah University of Science and Technology.

---

## ACADEMIC SERVICE AND PRESENTATIONS

- Peer Reviewer, *Journal of Biomedical Informatics* (Elsevier), 2025 – Present.
- Presenter, American Medical Informatics Association (AMIA) 2025 Annual Symposium, Atlanta, GA, USA.
- Organizer, International Conference on Computer and Information Technology (ICCIT) 2020; Organizer, Intra-AUST Programming Contest 2018; Advisor, Cefalo CodeFiesta 2022 Inter-University Programming Contest.
- Participant, ICECIT 2021 and IEEE TENSYP 2020, Dhaka, Bangladesh.

Citations: 200+ (Google Scholar). Last updated: September 2026.